KAMALESH SURESH KUMAR

BACKEND & SYSTEMS SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in backend architecture, systems programming, and high-efficiency deterministic routing engines. Demonstrated hands-on experience in TypeScript, Node.js, Python, C/C++, and SQL with focus on modular API design, low-latency client-side query evaluation, and systems security. Proven ability to design reliable, decoupled services with strict data typing and zero-downtime automated workflows.

TECHNICAL SKILLS & CORE COMPETENCIES

Languages: TypeScript, Python, C/C++, JavaScript, SQL
Backend/Runtimes: Node.js, Express, Linux / Unix CLI, Shell Scripting
Storage & Networking: SQL, REST APIs, Optical Wireless Protocols (OWC)
DevOps: GitHub Actions, Git, Vite Build Tooling, Automated Testing

Key Competencies: Backend & Systems Architecture * API Design & Intent Routing * Data Structures & Complexity Optimization * Relational Data Modeling & SQL * Memory Management & Embedded Logic * Automated CI/CD Validation Gates

KEY ENGINEERING PROJECTS

Rule-Based Conversational Router & Knowledge Dispatcher | *TypeScript, Natural Language Processing, Intent Routing, Algorithms*
- Engineered a deterministic intent resolution engine and contextual dispatcher routing visitor queries to spatial coordinates and typed records in sub-10ms.
- Architected clean interface boundaries decoupling knowledge data from UI layers, providing zero-runtime-cost query execution.

Li-Fi Penetration Testing & Optical Security Protocols | *Python, C/C++, Optical Wireless Communication, Signal Processing, Linux*
- Developed low-level testing harnesses in Python and C/C++ simulating packet interception and optical signal attenuation on Li-Fi RX 2.0/3.0 hardware.
- Implemented validation routines distinguishing ambient optical noise from exploit injection vectors, boosting vulnerability detection accuracy by over 40%.

Automated CI/CD Knowledge & Build Pipeline | *GitHub Actions, Node.js, Bash, YAML*
- Engineered automated deployment workflows running static analysis, compiler gates, and deterministic bundle generation.

EXPERIENCE & LEADERSHIP

Technical Trainee / Intern — Infosys SpringBoard *(2023 — 2024 | Virtual / India)*
- Completed advanced training in enterprise backend paradigms, cloud architectures, and algorithmic data processing.

Software Development Intern — Viyuga Innovations *(2024 | India)*
- Built and reviewed modular software routines, adhering to strict enterprise code quality and test plans.

Event Team Member / Coordinator — IEEE Computer Society *(2023 — Present | India)*
- Coordinated competitive programming hackathons, student workshops, and technical symposiums.

EDUCATION

B.E. in Computer Science and Engineering — Undergraduate Engineering Program *(2022 — Present | India)*
- Core Coursework: Systems Architecture, Data Structures, Algorithms, Operating Systems, Computer Networks, Database Systems, Network Security.

CERTIFICATIONS & VERIFIED HONORS

- Enterprise Software & Cloud Computing Certification — Infosys SpringBoard (2023 — 2024)
- Li-Fi Penetration Testing Demonstration & Law Enforcement Briefing — Cybercrime Police Department (2024)
- NCC Naval Wing Cadet Leadership & Service — National Cadet Corps (2022 — Present)